GROUP5.

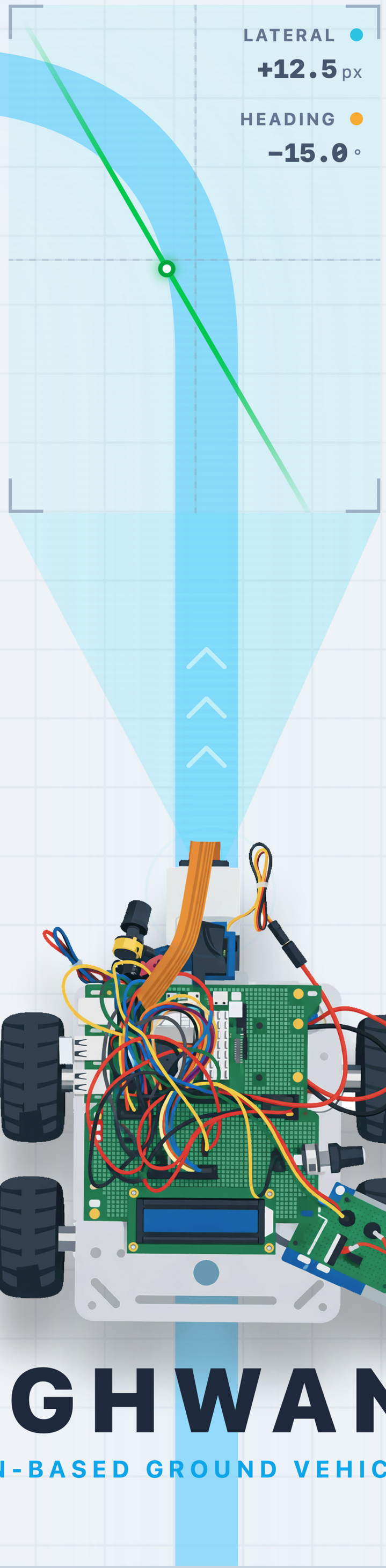
APPLIED ENGINEERING PROJECT

The University of Nottingham

VISION SYSTEM

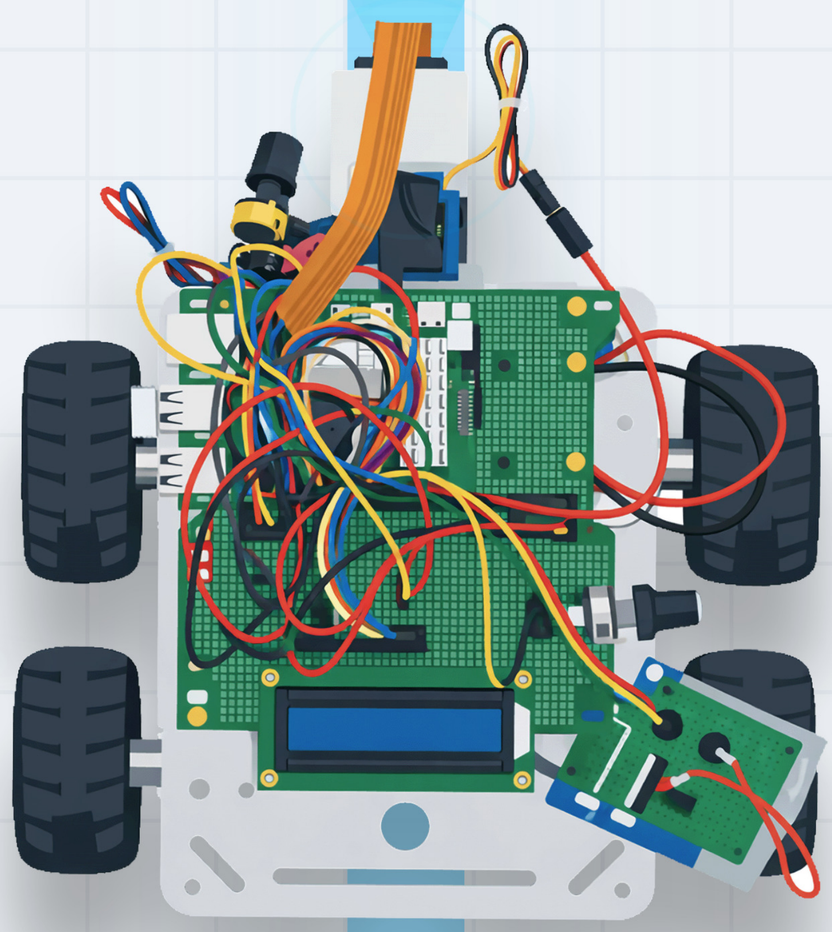
- Real-time lane detection via **cv::fitLine**
- Line loss memory
- Lateral offset & heading angle computation

ROI ANALYSIS



SYSTEM SPECIFICATIONS

SOFTWARE	C++ / OpenCV
CONTROLLER	Raspberry Pi
VISION	Camera Module 3
ACTUATION	1-DOF SG90
DRIVER	UART Driver Board
CHASSIS	4WD Differential
POWER	18650 × 2



GROUP MEMBERS

Luxuan SUN
Junyang ZHANG
Zichao MENG
Zihao ZHAO

NIGHWAN

AUTONOMOUS VISION-BASED GROUND VEHICLE